

Day 5

Modeling across domains and modalities

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'Levels of linguistic analysis'

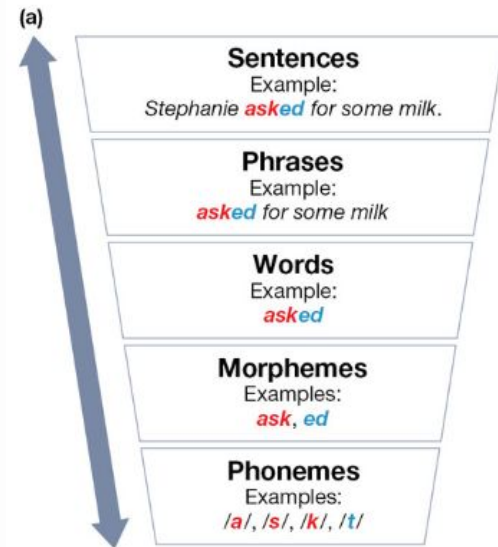
The Ohio State University Linguistics

Department. *Language Files: Materials for an*

Introduction to Language and Linguistics

- Chapter 1: Introduction
- Chapter 2: Phonetics
- Chapter 3: Phonology
- Chapter 4: Morphology
- Chapter 5: Syntax
- Chapter 6: Semantics
- Chapter 7: Pragmatics

Grison & Gazzaniga. *Psychology in Your Life*



Discrete levels?

- Phonetic detail affects lexical access (McMurray et al., 2002)
- Lexical knowledge biases ambiguous phoneme categorisation (Ganong, 1980)
- Prosodic boundaries cue syntactic parsing (Price et al., 1991)
- Children use a verb's syntactic frame to infer its meaning (Gleitman, 1990)

More broadly: usage-based acquisition theories, construction grammars

Discrete levels?

- Symbolic representations (at any level of granularity) are abstractions inferred from highly variable, noisy, rich perceptual signals
- In modeling, boundaries may also only exist in our descriptions of models, not necessarily in the models' representations

Modeling across levels

Interesting question: what happens *across the level boundaries*?

Hypothetical scenarios:

- A speech model that prefers grammatical sentences with no parser is evidence that syntax-sensitive behavior can arise from continuous, distributional representations.
- A model producing lexical behavior with no lexicon is an argument that the lexical level need not be separately represented to get lexical behavior

Modeling across levels



Difficulties with non-native spoken words

Non-native phonetic perceptual space:

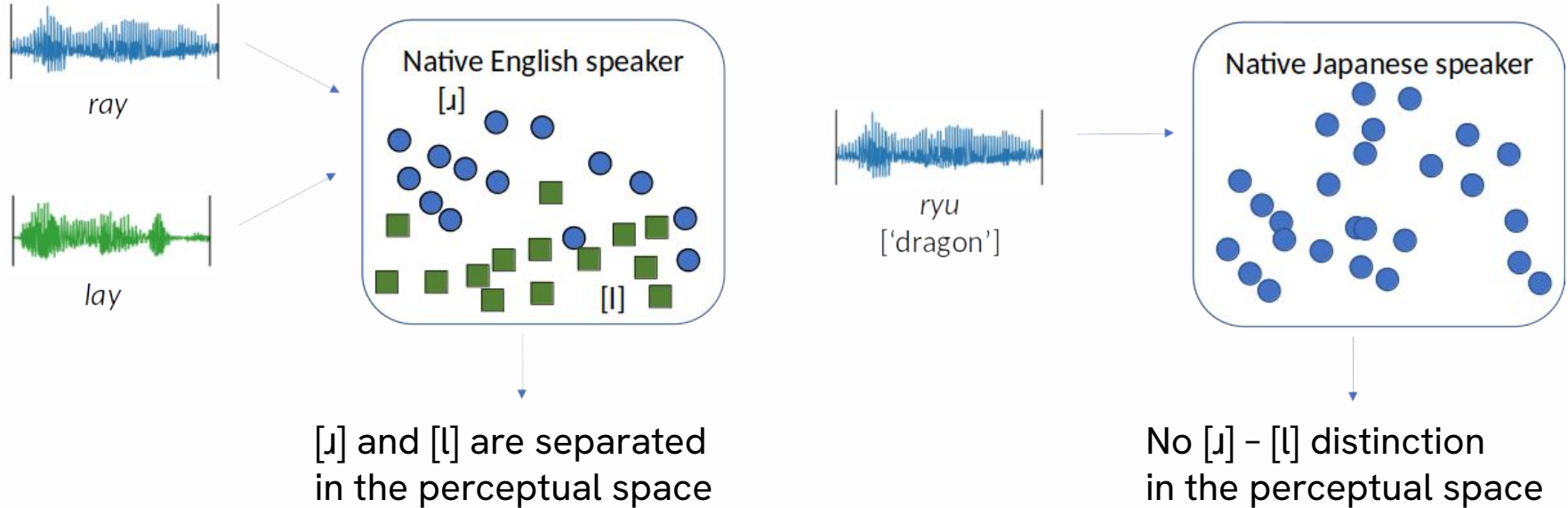
- Japanese speakers tested in English (MacKain et al., 1981)
- Confuse words such as *rock* and *lock*.
- An English phone contrast [ɹ]-[l] novel to Japanese speakers.

Imprecise phonological encoding of words:

- English speakers tested in Russian (Cook et al., 2016)
- Confuse *moloko* ['milk'] and *molotok* ['hammer'].
- Russian phone contrasts are **familiar** to English speakers.

Idea: show that a *phonetic* model can predict the “*phonological*” result.

Non-native phonetic perceptual space



Difficulties with non-native spoken words



moloko
['milk']



molotok
['hammer']

Native Russian speaker

[məɫʌ'ko]

[məɫʌ'tok]

Very precise
word phonology



moloko
['milk']



molotok
['hammer']

Native English speaker

[məɫʌ'ko]

[məɫʌ'tok]

Imprecise
word phonology

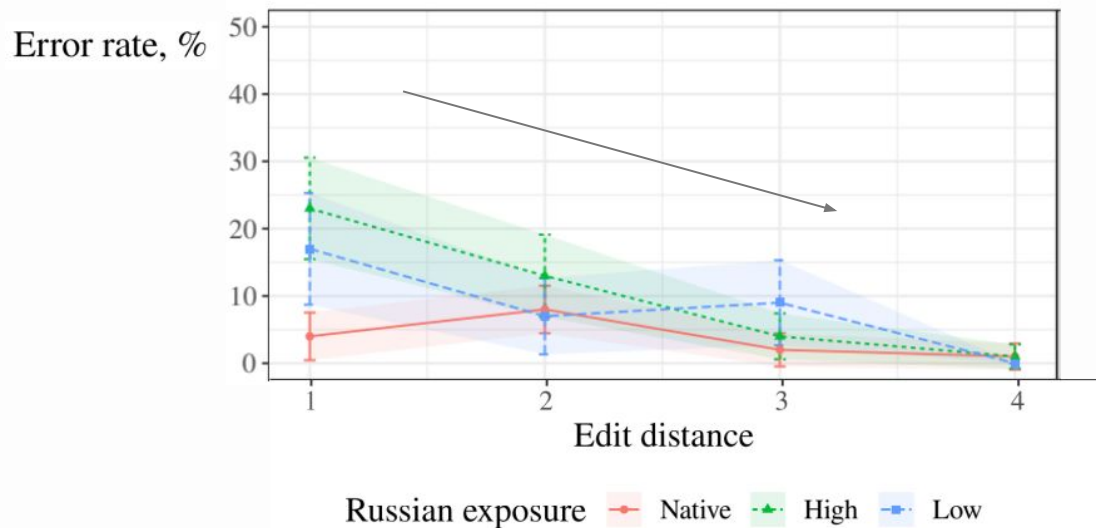
Study outline

- Matusevych et al. (2021): Can a phonetic model reproduce “phonolexical” difficulties?
 - The difficulties may have to do with phonetic perceptual spaces instead.
- A computational model of phonetic learning.
 - Train on speech from one or two languages.
 - Test on word discrimination (*moloko* – *molotok*).
 - Does non-native model show difficulties with spoken word processing?

Human participant data

- Cook et al. (2016): two groups of participants:
 - Native Russian speakers proficient in English
 - Native English speakers learning Russian
- Translation judgement task:
 - Hear Russian *moloko* ['milk'] → see English *hammer*.
 - Is *hammer* a translation of *moloko*
 - (No, it's a translation of *molotok*)
 - Key manipulation: phone edit distance between the two Russian words.

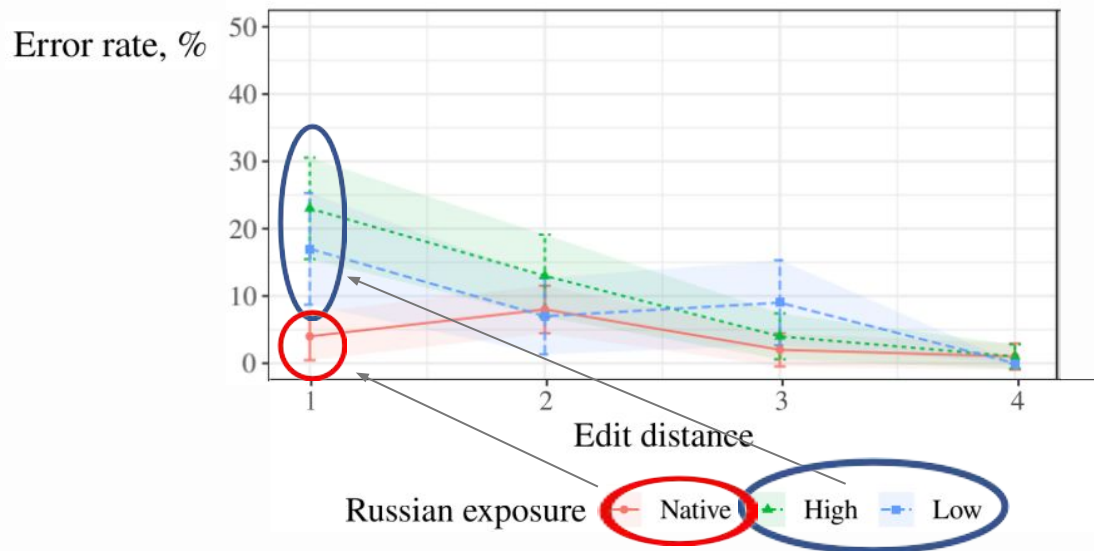
Human participant data



Larger edit distance –
lower error

Data from Cook et al. (2016)

Human participant data



Lower error in
native speakers

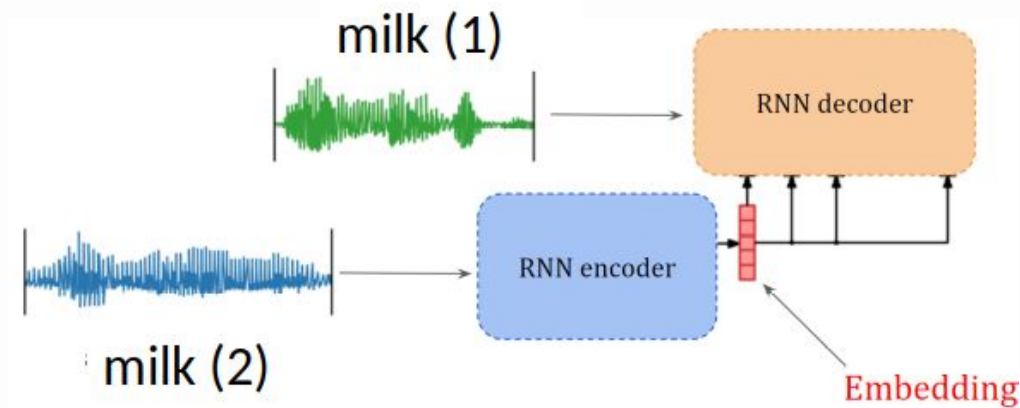
Data from Cook et al. (2016)

Our computational simulations

- Correspondence autoencoder recurrent neural network (CAE-RNN):
 - Proposed for speech engineering applications. (Kamper, 2019)
 - Used to simulate infant phonetic learning. (Matusevych et al., 2023) - cf. Wednesday lecture
 - It is a model of phonetic perception, no explicit phonology

Our computational simulations

- Never exposed to phones in isolation.
- Does not store explicit phonological representations of words.
- Learns by reconstructing acoustic words, **no** contrastive loss.



Validating the model

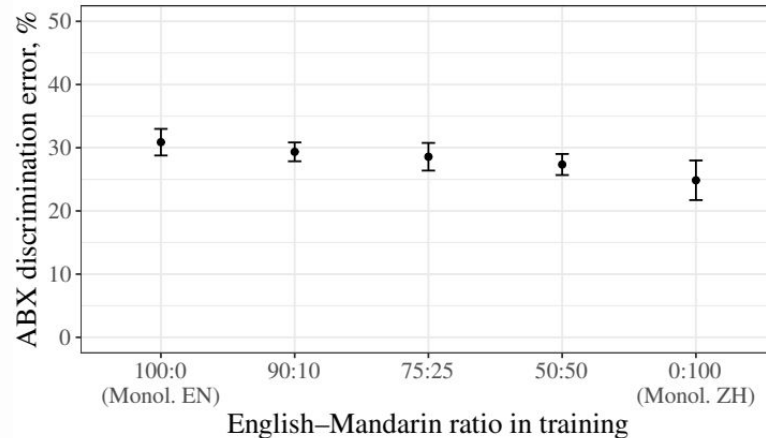
We want to model **word-level discrimination** in a **bilingual** model

So far: trained on **monolingual** data, predicting **phone** discrimination
(Matusevych et al., 2023)

1. Does our model work as a bilingual model?
2. Does our model work on word discrimination?

Validating the model

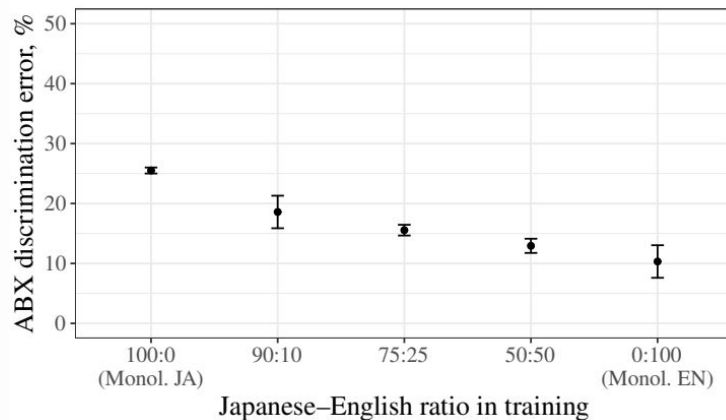
1. Does our model work as a *bilingual* model?
 - Test on a Mandarin phone contrast [ɕ] – [tɕ^h].
 - Discrimination patterns as in *bilingual* English-Mandarin infants. (Kuhl et al., 2003)



Matussevych et al. (2021)

Validating the model

2. Does our model work on *word* discrimination?
- Test on a word discrimination task: e.g., *rock* – *lock*.
 - Discrimination patterns as in bilingual English-Japanese speakers. (MacKain et al., 1981)

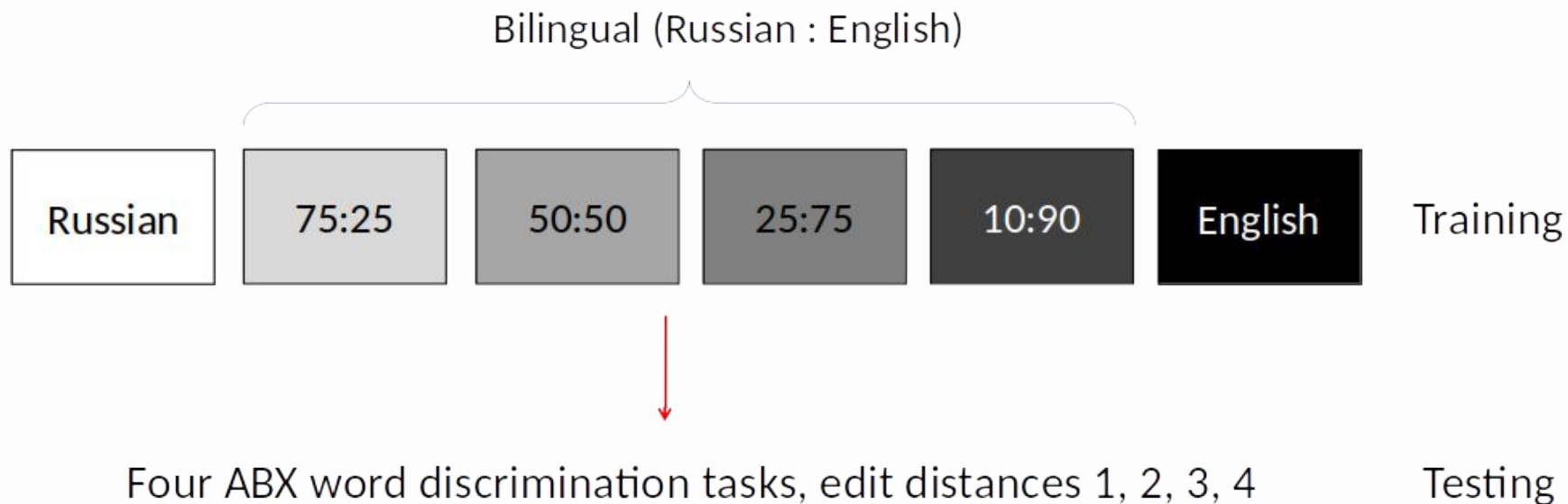


Matushevych et al. (2021)

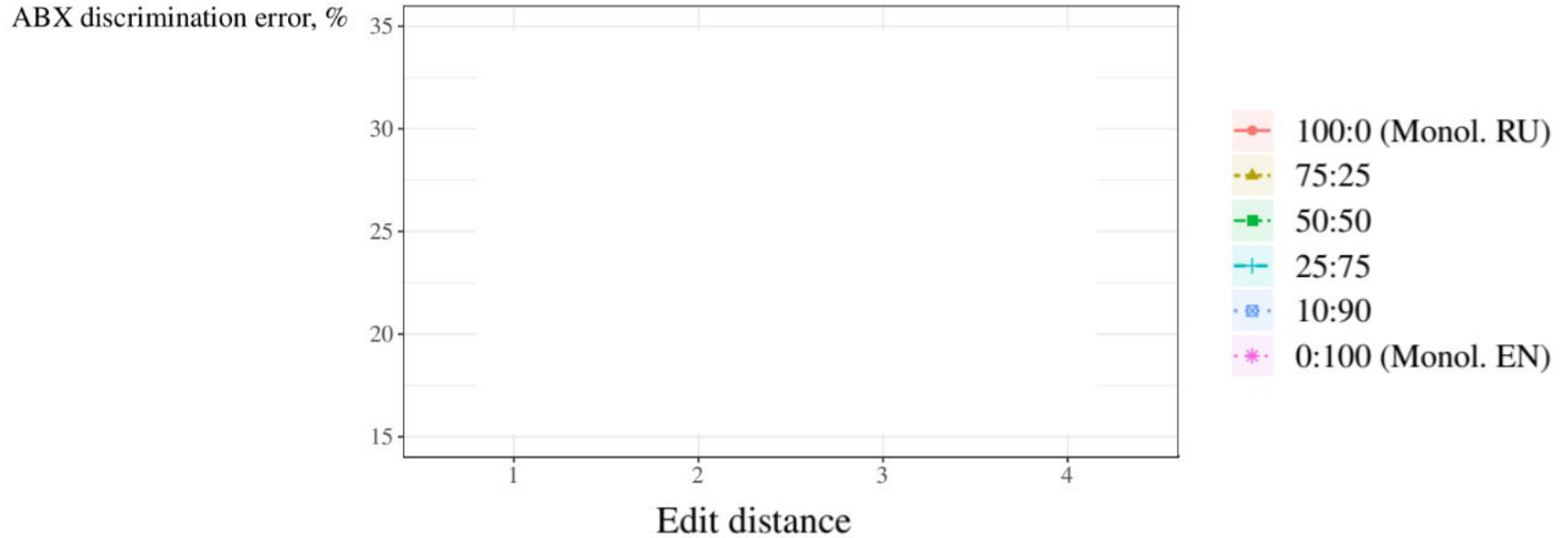
Main simulations

- Extract a set of Russian test words, encode them into model's representations.
- Select pairs that differ in 1, 2, 3, or 4 phones.
- Design 4 ABX discrimination tasks, one for each edit distance.
- An ABX triplet:
 - A: moloko (1)
 - B: molotok
 - X: moloko (2)
- X should be closer to A than to B.

Main simulations



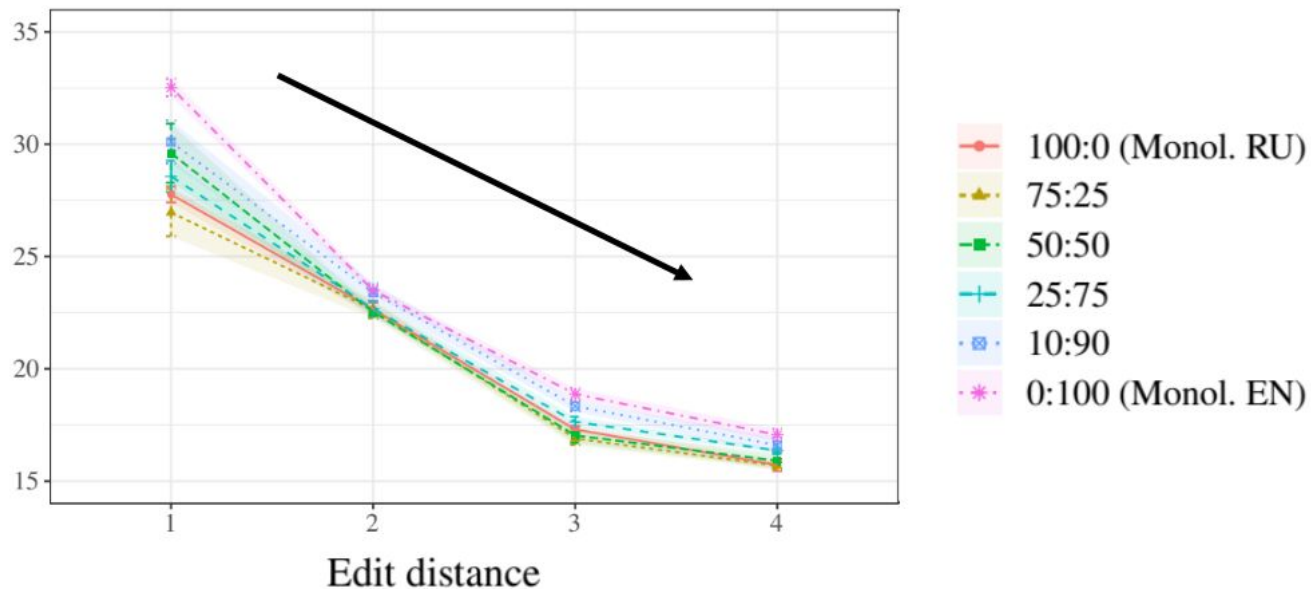
Main simulations



Main simulations

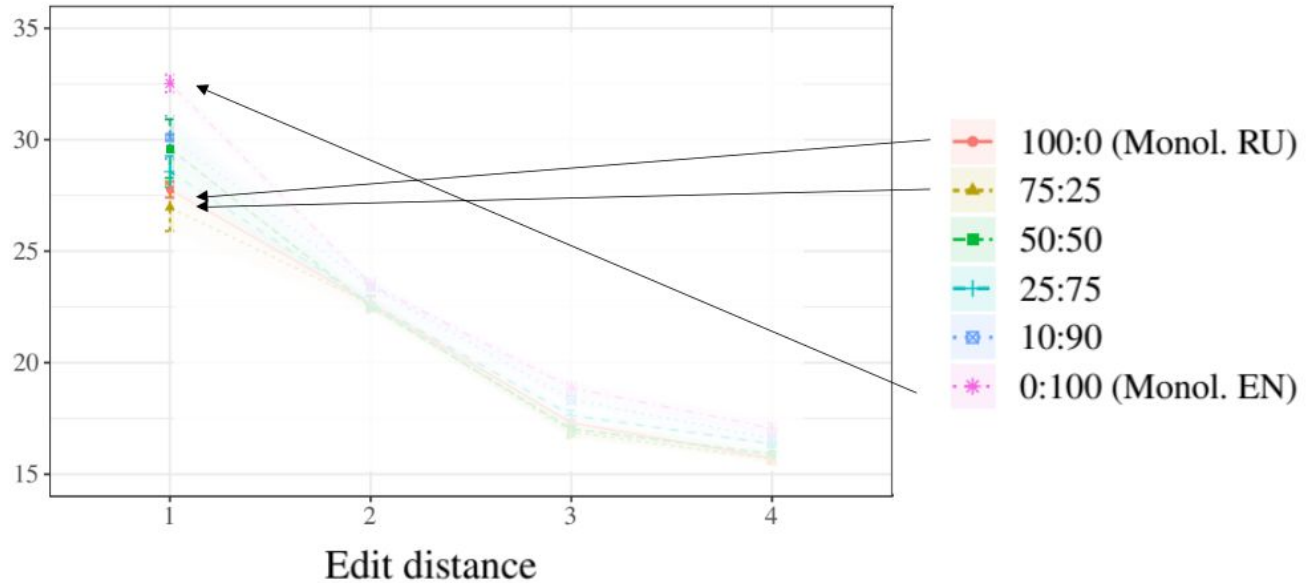
ABX discrimination error, %

Larger edit distance –
smaller error.



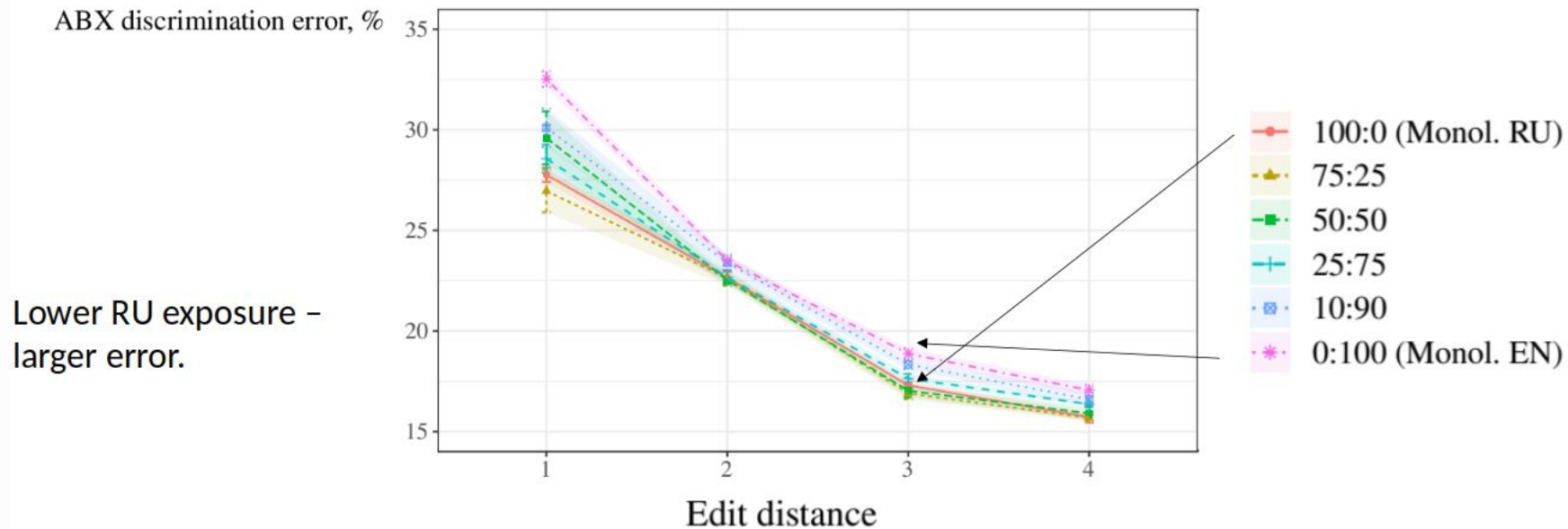
Main simulations

ABX discrimination error, %



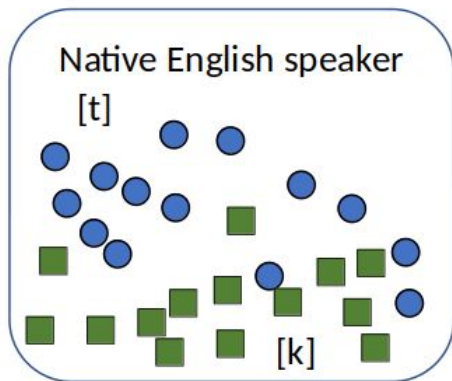
Lower RU exposure -
larger error.

Main simulations



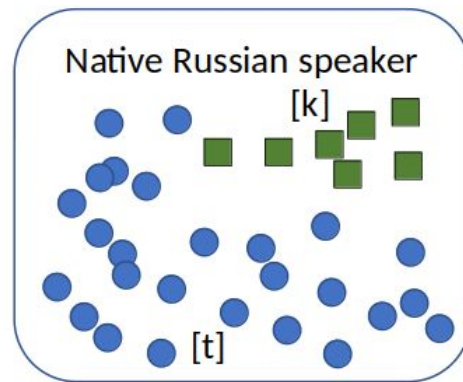
Phonetic perception vs. phonological encoding

- Our phonetic model shows patterns similar to human speakers in Cook et al. (2016)
- Difficulties in non-native spoken word processing can be partly explained by phonetic perception.



A speculative proposal

moloko - molotok



Modeling across levels

- This was an example of a model at the interface of phonetics (phonetic perception) and (lexical) phonology
- An example of joint phonetic and lexical learning: Lavechin et al. (2025)

Modeling across levels

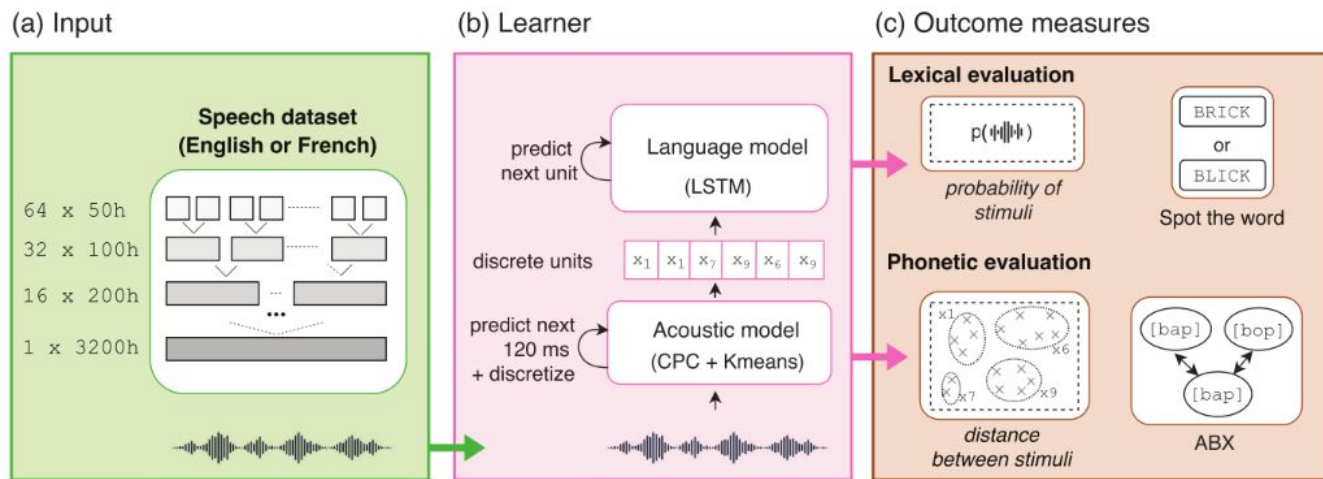


FIGURE 1 | Overall setup for the training and testing of STELA. (a) The audio environment of learners of different “ages” are modeled using audiobooks segmented and aggregated in increasingly larger sets matched for number hours and of speakers across two languages (See Table 1). (b) The learner is composed of an “Acoustic Model,” first trained with predictive coding and followed by a K-means algorithm returning discrete units and a LSTM “Language Model” trained to predict future units based on past units. (c) Outcome measures are obtained by modeling an ABX sound discrimination task at the (discretized) output of the Acoustic Level, and an auditory lexical preference task (Spot-the-Word) by using the ability of the Language Model to compute the probability of stimuli.

Lavechin et al. (2025)

Modeling across levels

- This was an example of a model at the interface of phonetics (phonetic perception) and (lexical) phonology
- An example of joint phonetic and lexical learning: Lavechin et al. (2025)
- Many modern neural networks - both S3M (self-supervised speech models) and LLMs - are operating across levels

Emergence of linguistic levels

- Lakretz et al. (2019): number and syntax units in LSTM
- Manning et al. (2020): syntax in BERT
- Pasad et al. (2024): words and meaning in S3Ms
- Goldstein et al. (2025): phonemes/morphemes/syntax/ semantics in LLMs (vs. dynamics in human brain)

Bottom line: linguistic structure emerges in NNs at various levels

Speech vs. text

Question: for a cognitive scientist, how do models trained on speech fundamentally differ from models trained on text?

Modeling across levels

How do we know what level a model knows?

- Behavioral analysis
 - *Does it behave as if it knows words?*
- Representation probing
 - *Is word-level information recoverable from activations?*
- Architectural modifications
 - *Does it form word units when designed to?*

Adding vision

Vision-language models

Chang & Bergen (2022): how neural language models acquire individual words during training?

- Extract ages of acquisition for 600+ words from the MacArthur-Bates Communicative Development Inventories, compare these to word learning in LMs
- Effects of concreteness, word length, and lexical class are very different in children and language models → interaction and sensorimotor experience are important

Vision-language models

- One way to add sensorimotor experience: vision data
- Good data for training models: image captioning

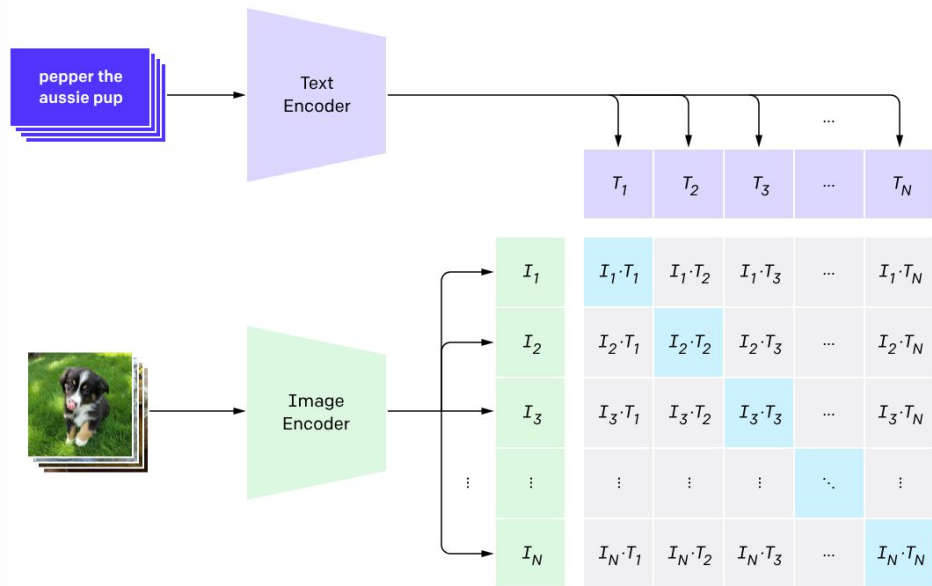
Popular vision-language models (VLM):

- CLIP (Radford et al., 2021)
- Flamingo (Alayrac et al., 2022)
- BLIP-2 (Li et al., 2023)
- LLaVA (Liu et al., 2023)

Vision-language models

Training CLIP:
predicting the correct
pairings in a batch of
(image, text) examples

1. Contrastive pre-training



Radford et al. (2021)

Vision-language models

Zhuang et al. (2024): does visual grounding make models learn words in a more naturalistic way?

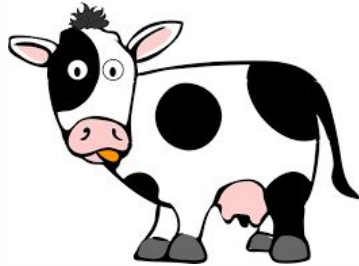
- Train models (GPT-2 and various VLMs) with and without visual supervision
- Test on tasks involving syntactic categories, lexical relations, semantic features, word similarity, brain activation alignment
- Visual grounding makes model's word learning more similar to human - *in a low-data regime* (with little distributional evidence from text alone)

Mutual exclusivity in child learning

Markman & Wachtel (1988)

Show me the dax!

Familiar object



Unfamiliar object

Computational models of ME

Learning to map words to their referents



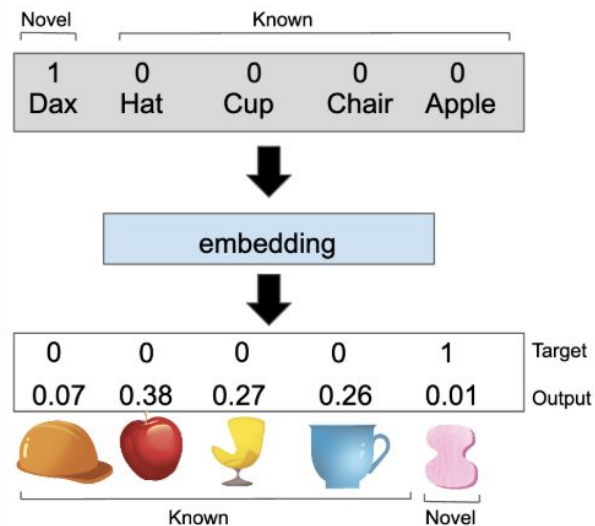
COW

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Computational models of ME

Gandhi & Lake (2020): synthetic data, simplistic representations, no ME effect



Computational models of ME

Gandhi & Lake (2020): synthetic data, simplistic representations, no ME effect

Gulordava et al. (2020): real image data, ME effect only when anti-synonymy is part of the loss function

Vong & Lake (2022): VLMs, ME bias non-existent relative to humans

Computational models of ME

- Mutual exclusivity did not automatically emerge in models:
 - Only if explicitly added to the model.
- Unfamiliar items are difficult to deal with:
 - Novel words are not grounded in the model's representation space.

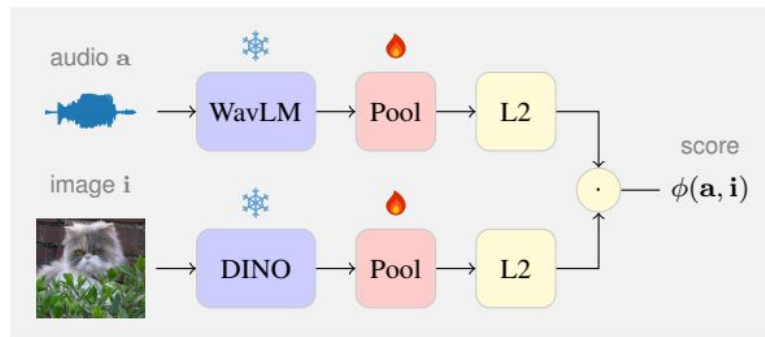
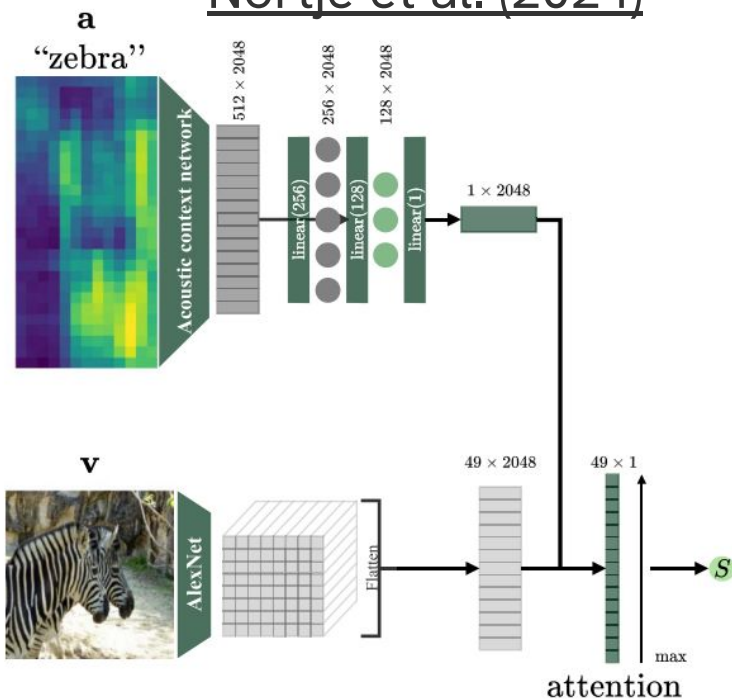
Visually grounded speech models

Learning to map acoustic words to their referents



Visually grounded speech models

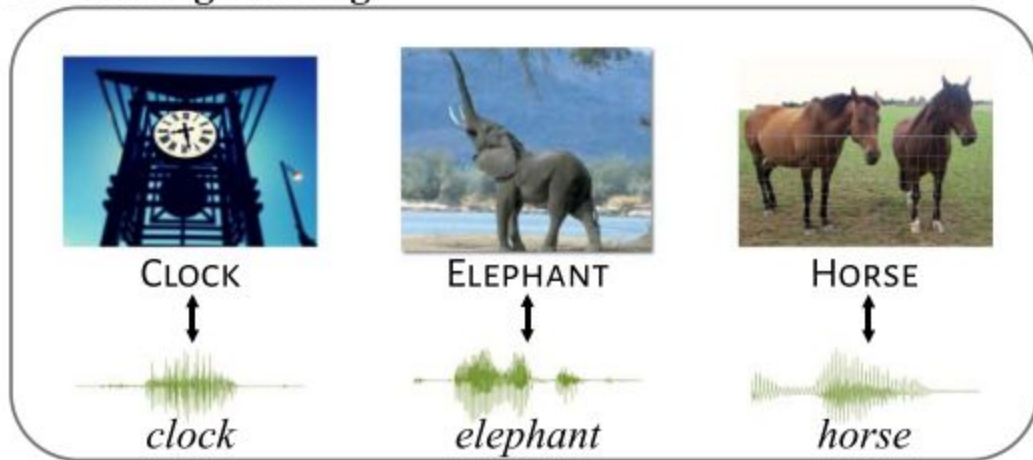
Nortje et al. (2024)



Oneata et al. (2025)

Visually grounded speech models

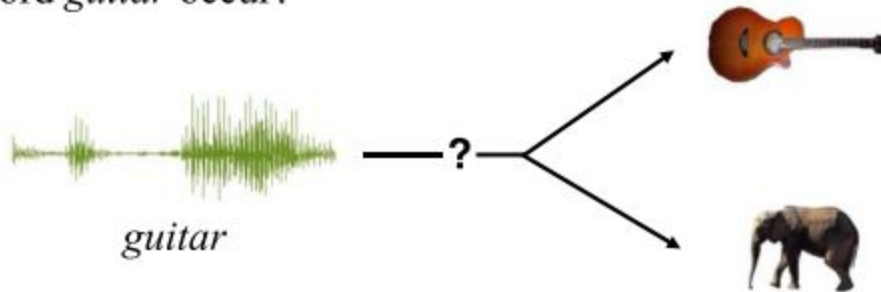
Given during training:



Training, from Nortje et al. (2024)

Visually grounded speech models

Test-time question: In which picture does the novel spoken keyword *guitar* occur?



Testing, from Nortje et al. (2024)

ME in visually grounded speech models

- Consistent mutual exclusivity bias in visually grounded speech models
 - Across different feature encoders
 - Across different losses
 - Across different languages
- Dual grounding (both in sound and vision) can help correctly predict human data using neural networks

Back to VLMs: text+vision

Vong et al. (2024): CVCL model

- Recorded one child's view from 6 to 25 months of age using a head-mounted video
- Used still frames and uttered words as training data



Back to VLMs: text+vision

Vong et al. (2024): CVCL model

- The model has learned word-referent mappings - good at selecting the target image given a word
- The mappings generalize: referents are recognized even on out-of-distribution stimuli (Konkle objects; Konkle et al., 2010)

Recap / discussion

Mon	What makes a computational model cognitive?
Tue	Architectural and data choices
Wed	Learning representations
Thu	Multilingual learning
Fri	Modeling across domains and modalities

Useful models from practicals:

- Pythia (LLMs, open-source data)
- Wav2vec2 (a large S3M)
- B-GPT (smaller bilingual LMs)
- CVCL (VLM, 1 child)
- CLIP (large VLM)

Your questions / thoughts?

Agree / disagree?

- Cognitive plausibility is not a property of a model, but of a model-claim pair.
- A model that performs well but has no clear failure criterion is scientifically uninteresting.
- Behavioral fit without representational fit is only curve-fitting.
- LLMs trained on internet-scale data are irrelevant to theories of child language acquisition.
- Shared multilingual representations are an engineering convenience, not a cognitive hypothesis.
- Linguistic levels are artifacts of analysis, not necessarily units of cognitive representation.

Thanks!

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